

ID:

Name:

**Brac University**

Semester: Summer 2025

Course Code: CSE251

Electronics Devices and Circuits

Set

A

Assessment: *Midterm Exam*

Duration: 1 hour 30 minutes

Date: July 26, 2025

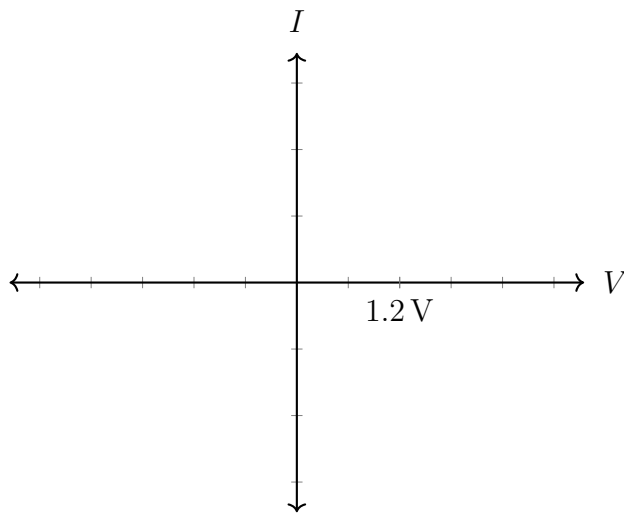
Full Marks: 60

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ All **5 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

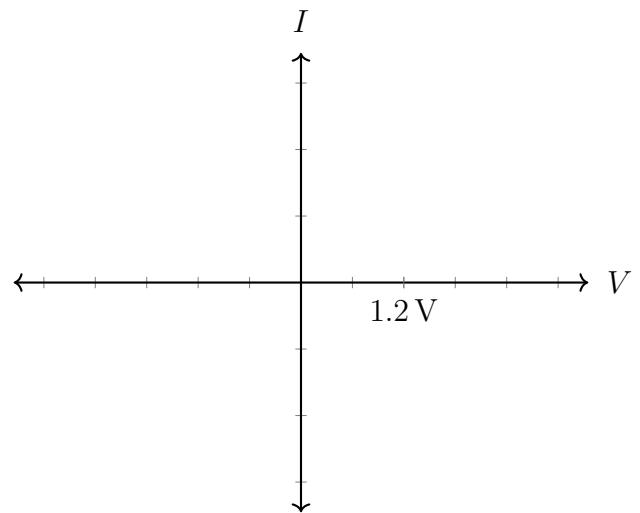
■ Question 1 of 5**[CO1] [10 marks]**

- (a) **[3 marks]** List three ideal characteristics of an operational amplifier (op-amp).
- (b) **[3 marks]** If V^+ and V^- denote the voltages at the non-inverting and inverting input terminals, respectively, is the condition $V^+ \approx V^-$ always valid for an op-amp? If not, under what scenario is this approximation valid? What is the condition called when either V^+ or V^- is 0 V while this approximation holds?
- (c) **[4 marks]** For a GaAs diode with a cut-in voltage of $V_{D0} = 1.2$ V, sketch the I-V characteristics for the following two models:
- Constant voltage drop model
 - Constant voltage drop plus resistance model

Label significant regions on both plots.



Grid for (i)



Grid for (ii)

■ Question 2 of 5**[CO3] [6 marks]**If x , y , and z are Boolean inputs and f is the output,

- (a) **[3 marks]** Design a circuit using diode logic to implement the following Boolean function:

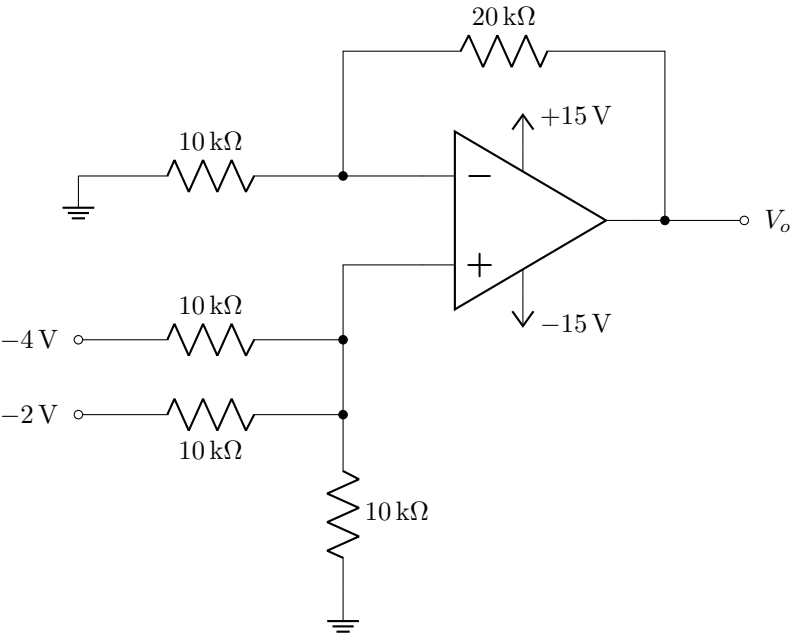
$$f = x \cdot y + z$$

- (b) **[3 marks]** Given the input voltages $x = 3.3$ V, $y = 3.0$ V, and $z = 2.7$ V, determine the output voltage of the logic circuit designed in (a). Assume a supply voltage of 5 V, where applicable. Also, assume that the diodes have a cut-in voltage of $V_{D0} = 0.7$ V and negligible resistance compared to other components in the circuit.

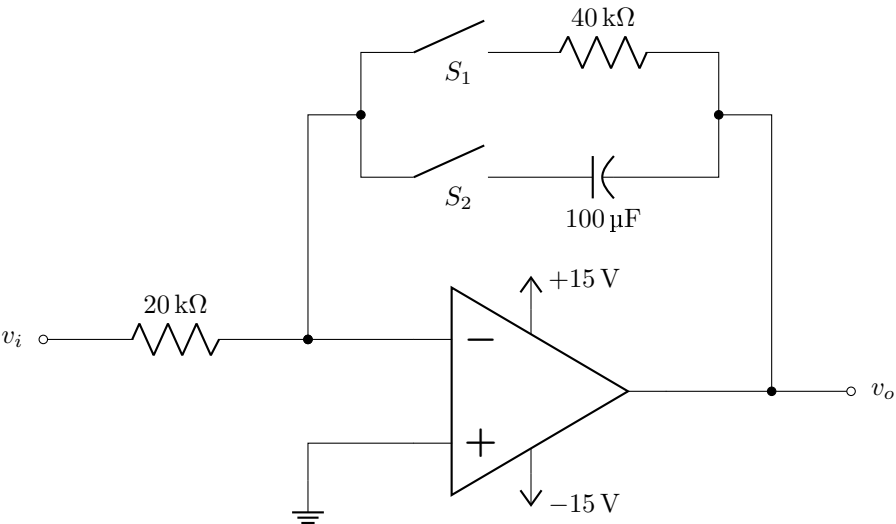
■ Question 3 of 5

[CO2] [16 marks]

(a) [10 marks] Analyze the circuit shown below and determine the voltage V_o .

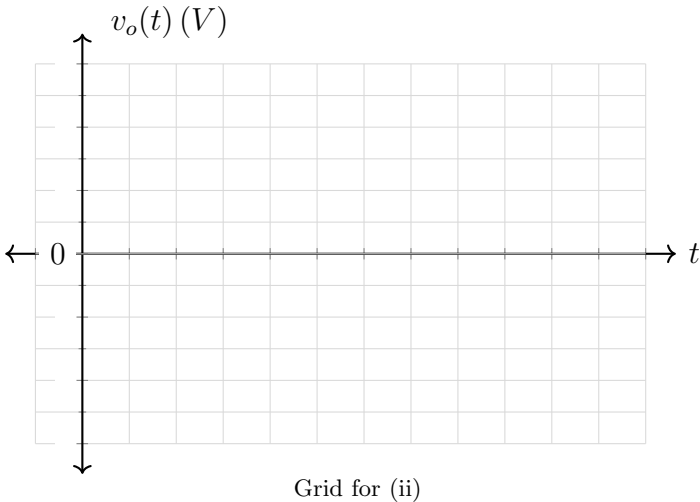
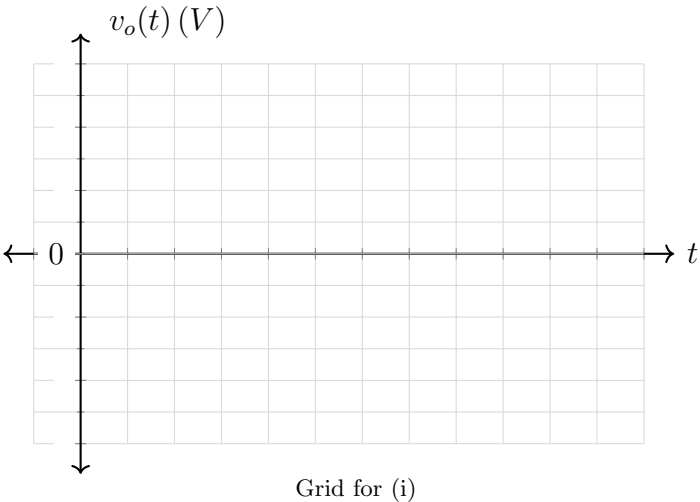


(b) [6 marks] In the following circuit, v_i is a sinusoidal voltage given by $4\sin(t)\text{ V}$.



Approximately sketch v_o versus t on the blank grids provided below when:

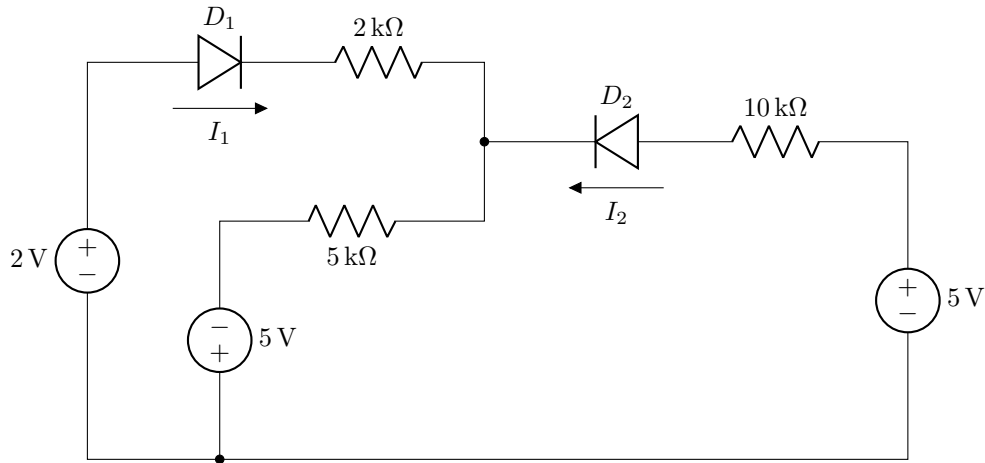
- (i) switch S_1 is closed and S_2 remains open.
- (ii) switch S_2 is closed and S_1 remains open.



■ Question 4 of 5

[CO2] [10 marks]

For the circuit shown below:



- (a) [2 marks] Draw a non-loop alternative representation of the circuit.
- (b) [8 marks] Assuming a cut-in voltage of 0.7 V for the diodes, determine whether diodes D_1 and D_2 are conducting or not. Also, calculate the power dissipated by each diode.

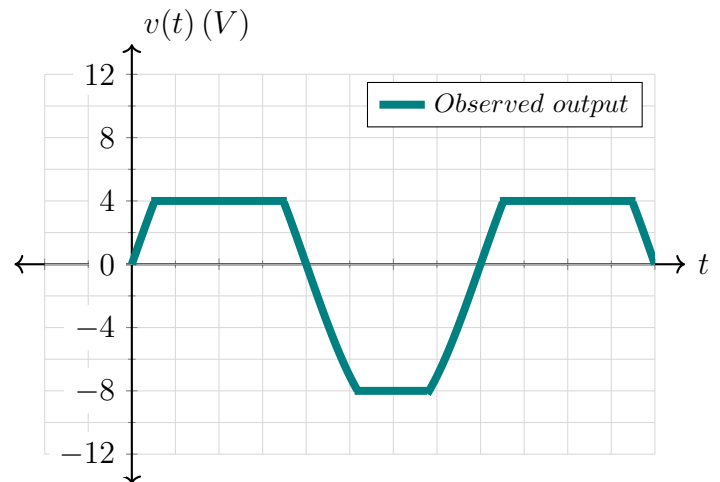
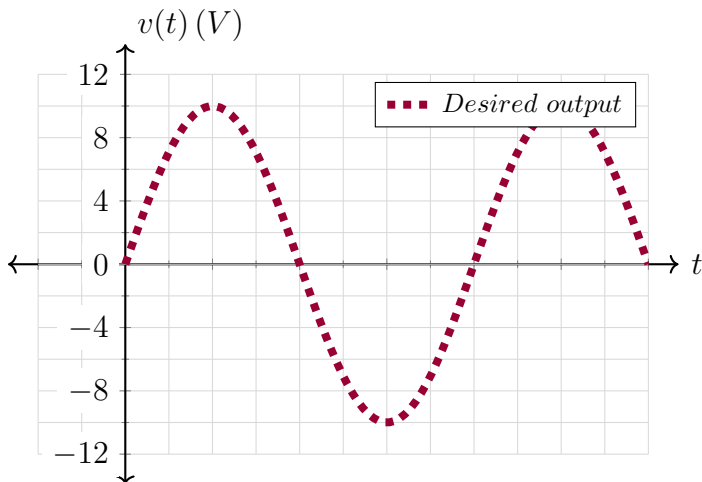
■ Question 5 of 5

[CO3] [18 marks]

- (a) [8 marks] Implement the following function using op-amp circuits. Here, x , y , and z are analog voltage inputs, and f is the voltage output.

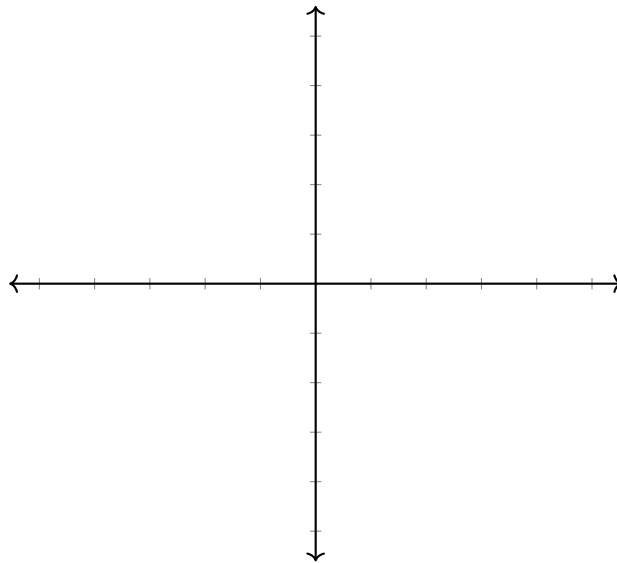
$$f = -5 \frac{d^2}{dt^2} x - \frac{5}{3} y + 7 \int z dt$$

- (b) An amplifier circuit was designed to amplify a weak sinusoidal voltage signal from a sensor, given by $2 \sin(t)$ V, with a desired output waveform as shown in the left plot below. However, when the output voltage of the amplifier was observed on an oscilloscope, the waveform appeared as a distorted sinusoid, similar to the one shown on the right.



- (i) [4 marks] Draw the amplifier circuit that produced the distorted sinusoidal output shown above, specifying all relevant component values.

- (ii) [3 marks] Sketch the voltage transfer characteristic (VTC) of the amplifier designed in part (i) on the grid provided below. Label the axes appropriately, and indicate any relevant details on the plot.



- (c) [3 marks] Design an op-amp circuit that follows the VTC shown below. v_i and v_o represent the input and output voltages, respectively.

